

Assignment 7

1. Show that the Quaternion group is not a semidirect product of two proper subgroups.
2. Show that if p is a prime, a group of order $2p$ is either cyclic or the dihedral group.
3. Describe the (unique) non-abelian group of order 21 using generators and relations.
4. Show that any group of order 75 is a semidirect product of two proper subgroups.
5. Show that product of two solvable groups is solvable.